

VERSION HISTORY

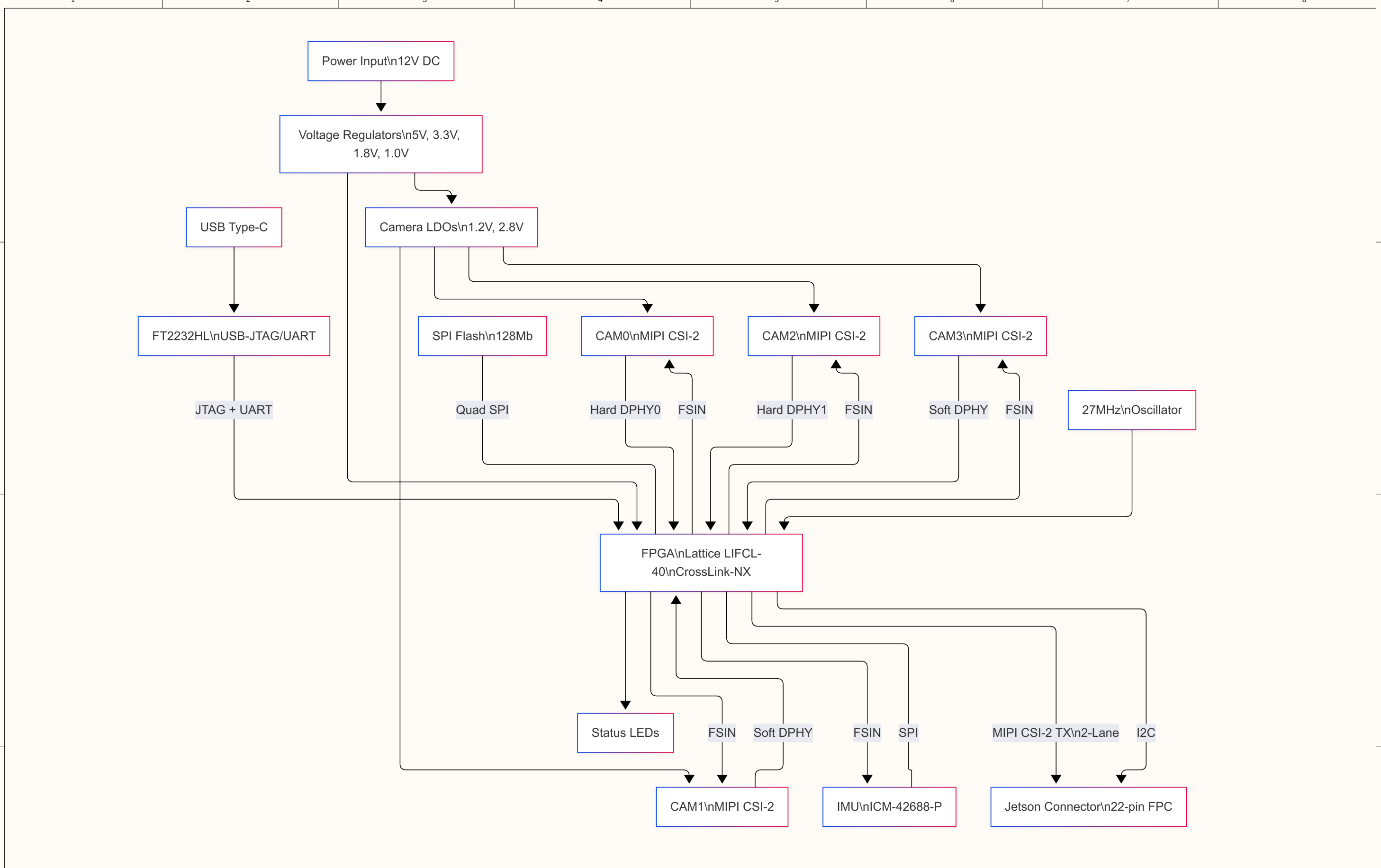
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Design Notes

Revision	Description	Date	Drawn By
R1	Initial Release	24.02.2026	Umit KAYACIK

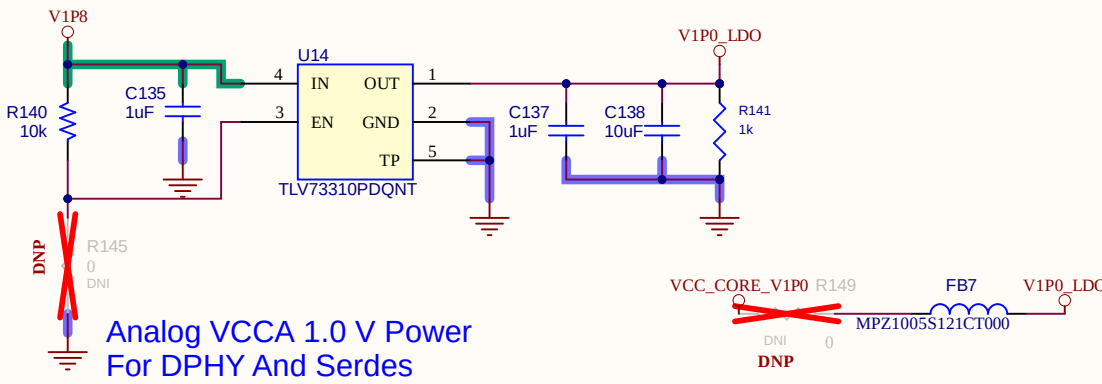
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		Umit KAYACIK	Revision:
	Son Güncelleme/Last Edit Date	Kontrol eden/Checked By	Boyut/Size: A3
	24.02.2026		Dil/Language: ENG
		Onaylayan/Approved By	Varyant/Variant:
			Variant of HW_CHRONOS_R1



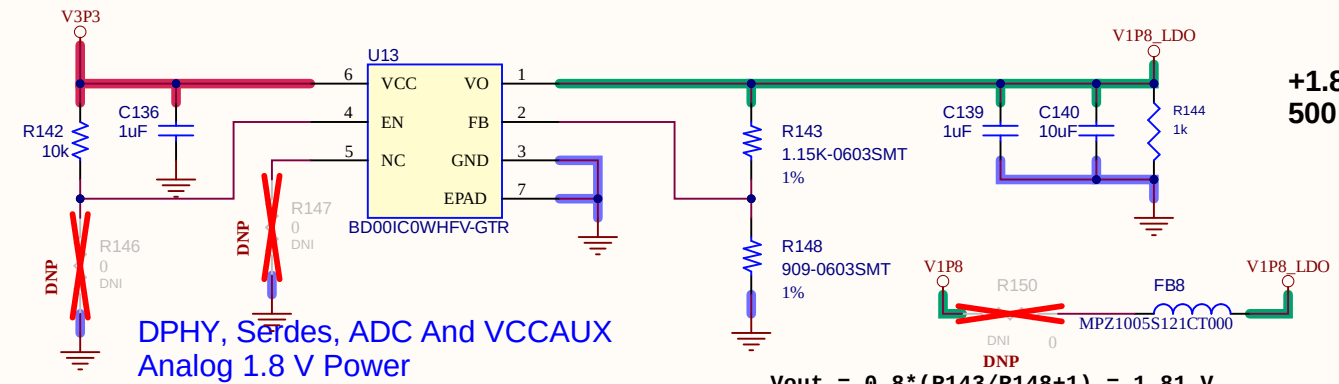
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	24.02.2026		Dil/Language: ENG
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			Variant of HW_CHRONOS_R1

POWER REGULATORS

**+1.0 V
300 mA**

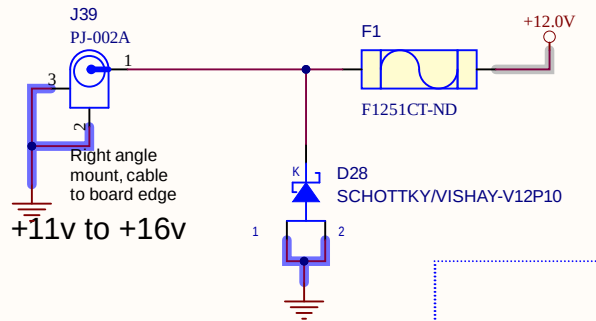


**+1.8 V
500 mA**

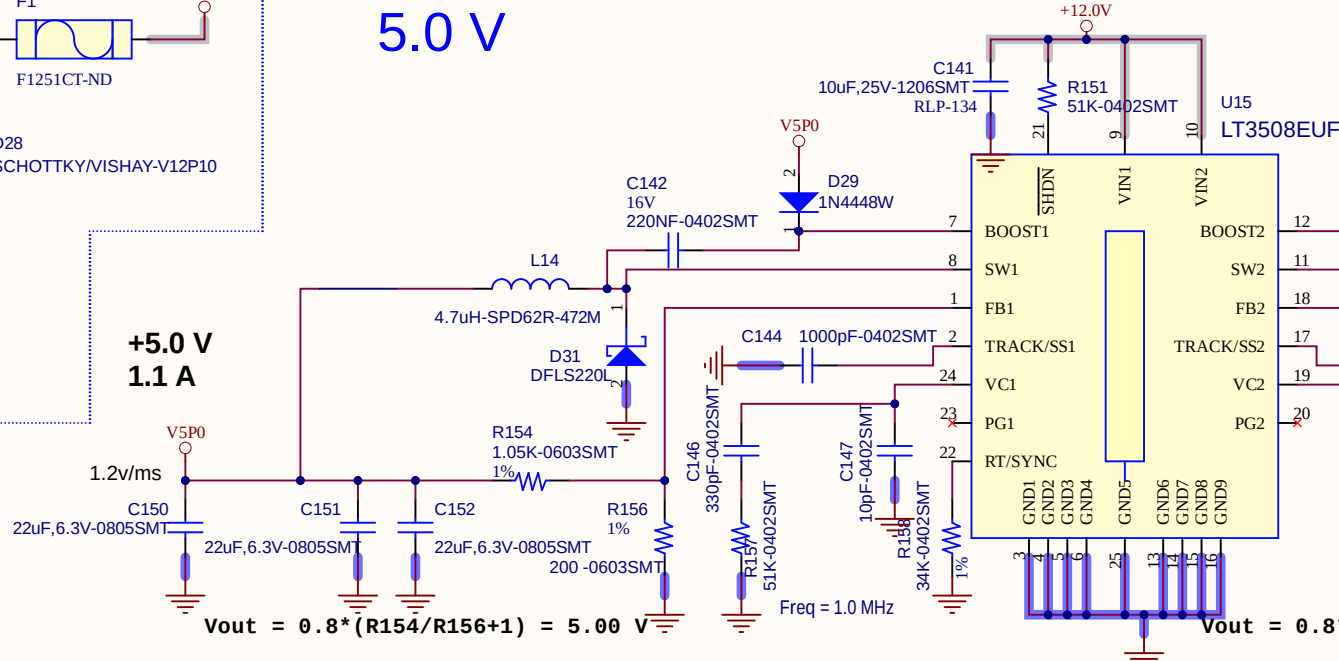


5.0 V

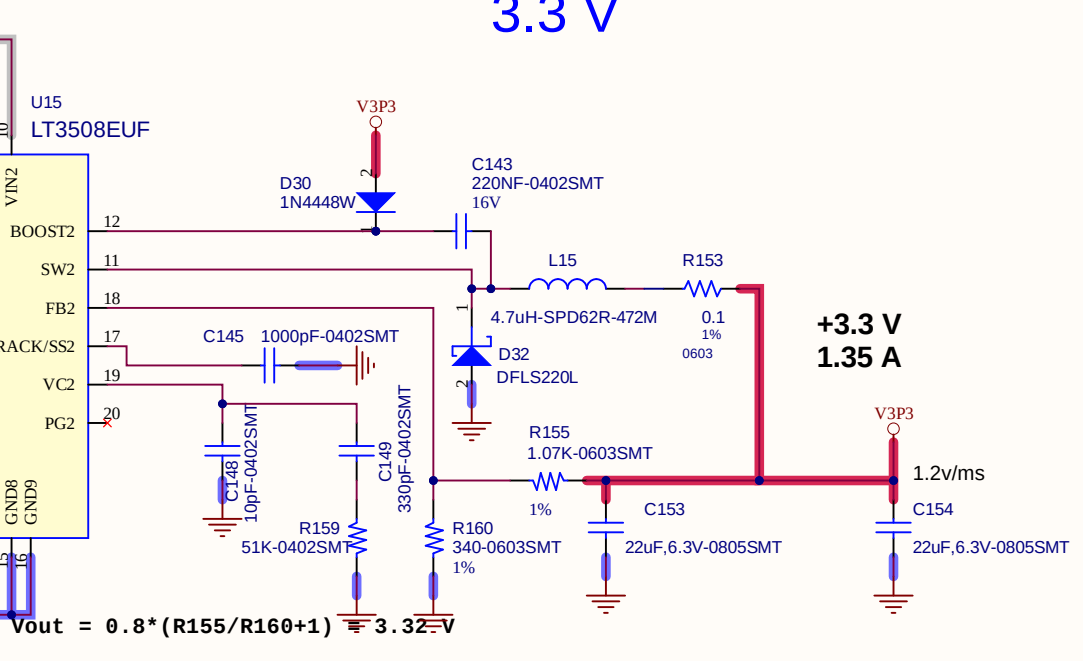
3.3 V



**+5.0 V
1.1 A**



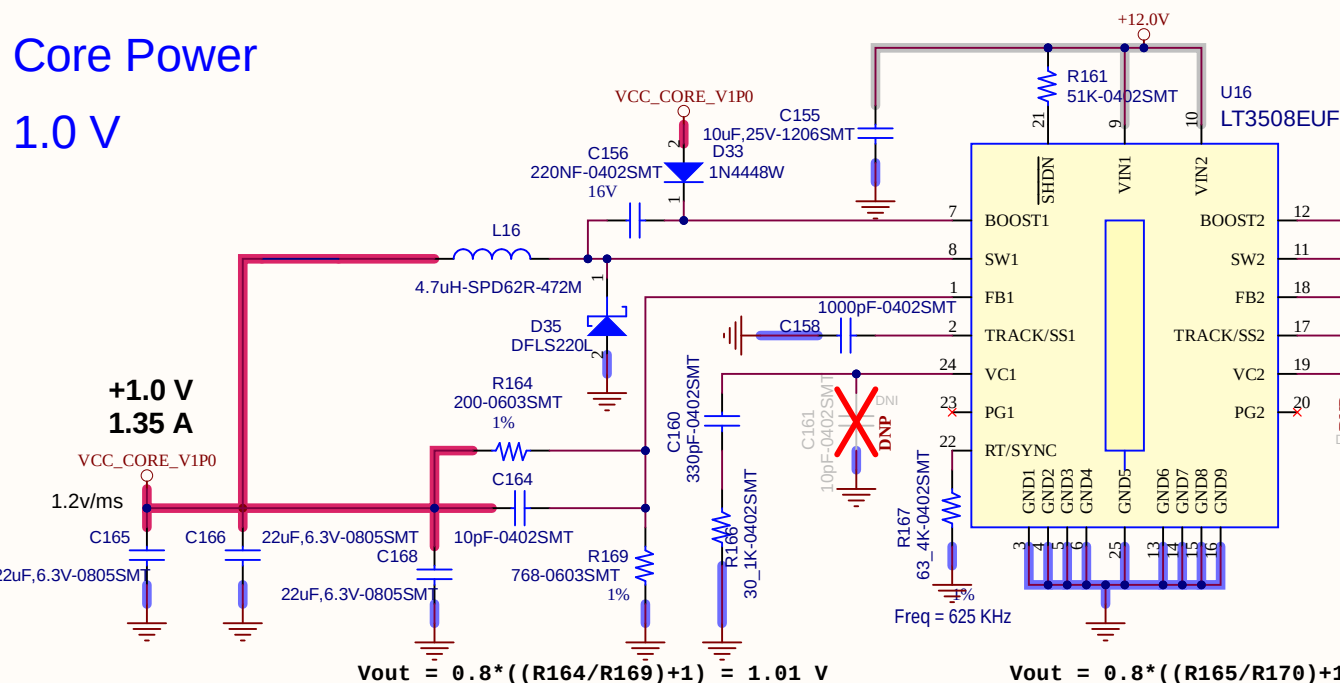
**+3.3 V
1.35 A**



Core Power

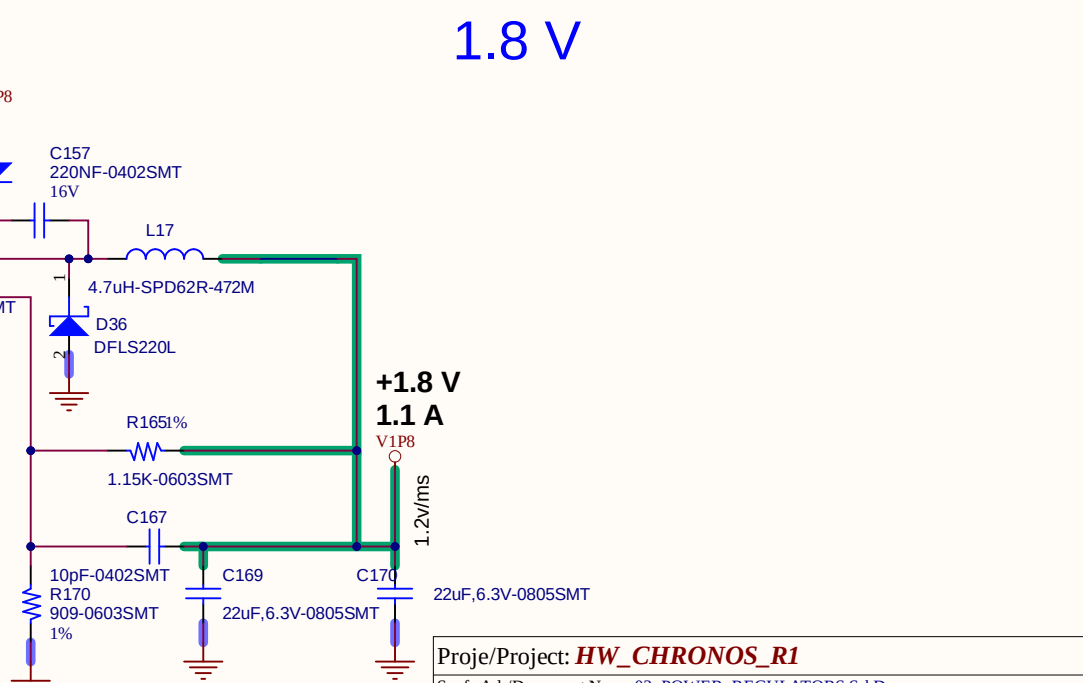
1.0 V

1.8 V



Vout = 0.8 * ((R165/R170)+1) = 1.81 V

**+1.8 V
1.1 A**



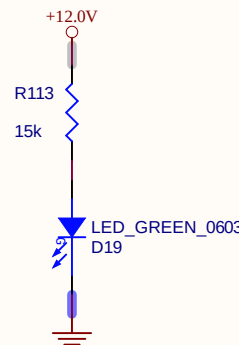
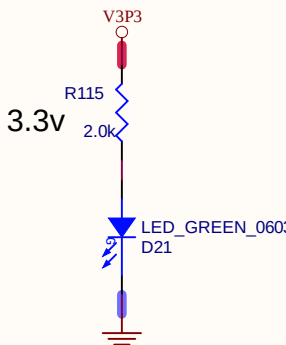
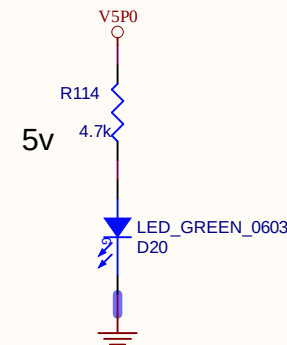
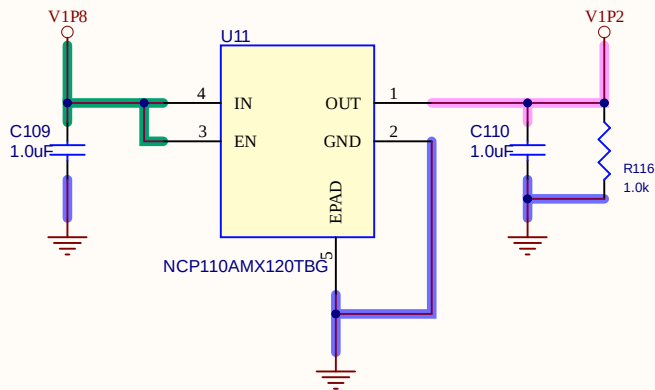
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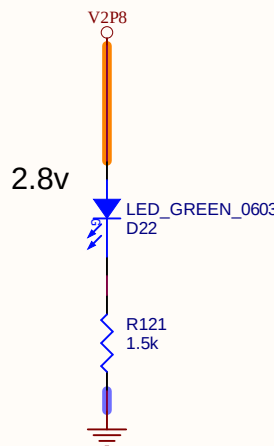
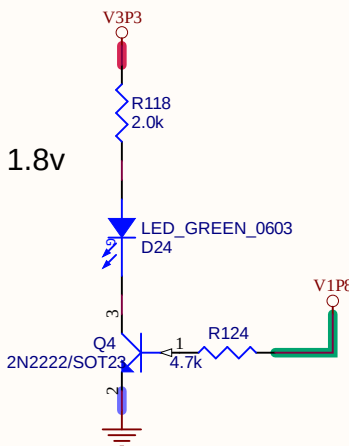
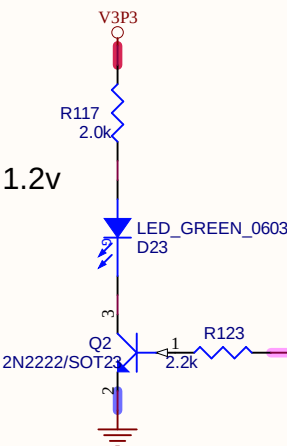
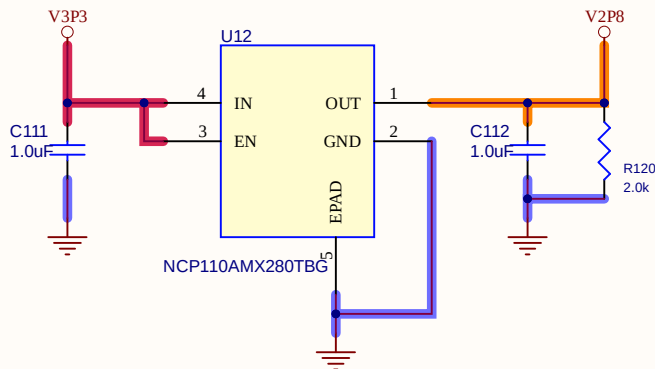
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	Son Güncelleme/Last Edit Date	Kontrol eden/Checked By	Revision: *
	24.02.2026	*	Boyut/Size: A3
		Onaylayan/Approved By	Dil/Language: ENG
		*	Varyant/Variant: Variant of HW_CHRONOS_R1

POWER CSI AND BANK POWER

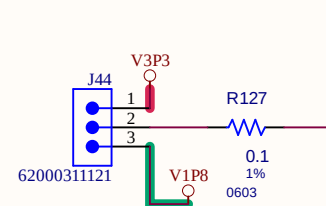
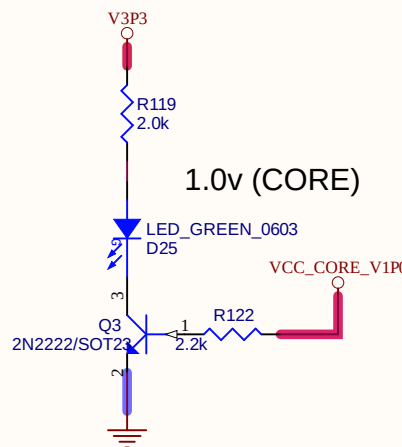
12VIN GOOD



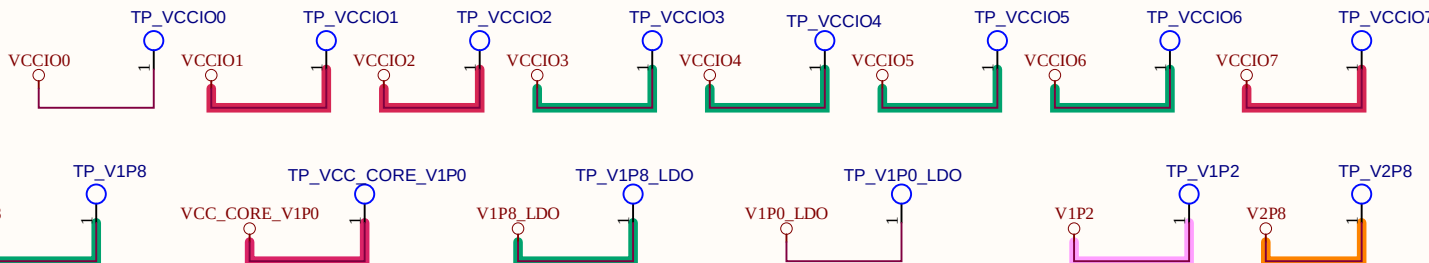
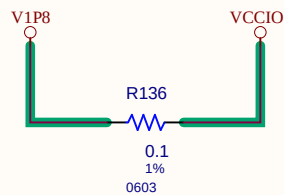
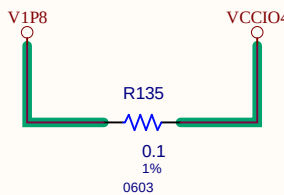
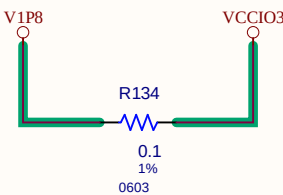
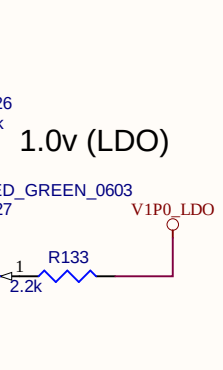
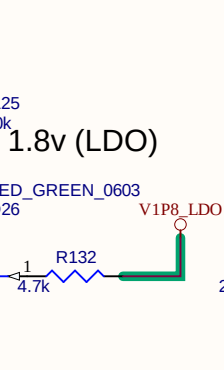
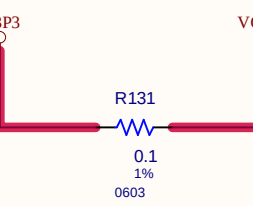
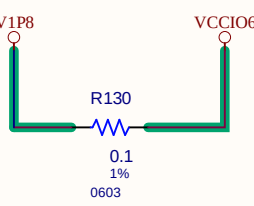
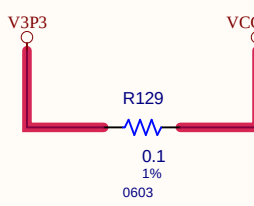
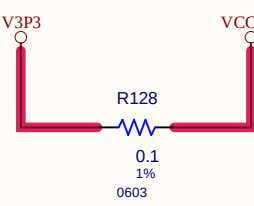
Group LEDs together



1.0v (CORE)

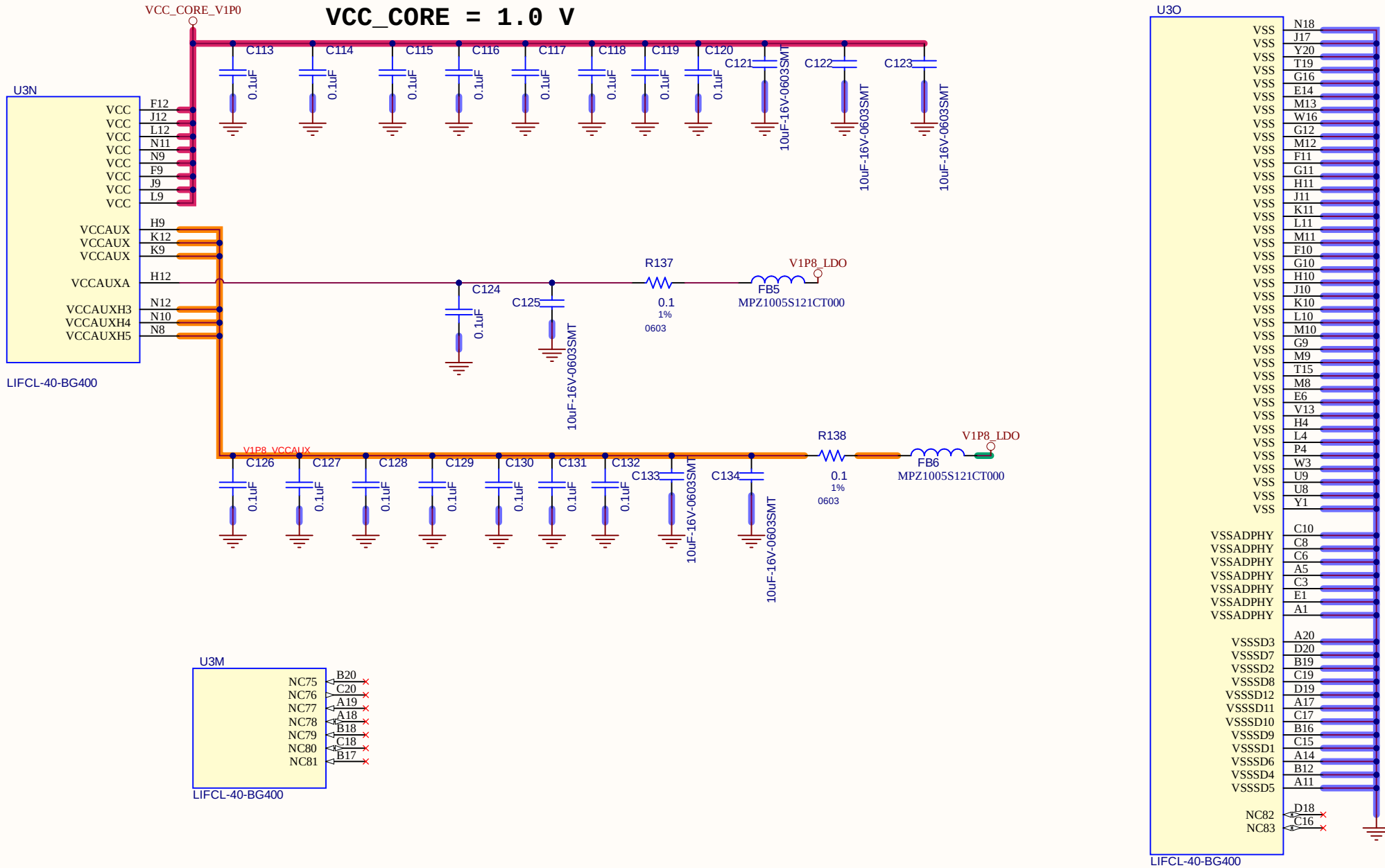


Flash Voltage Select
1-2 3.3V (Default)
2-3 1.8V (Note PMOD2 unavailable)

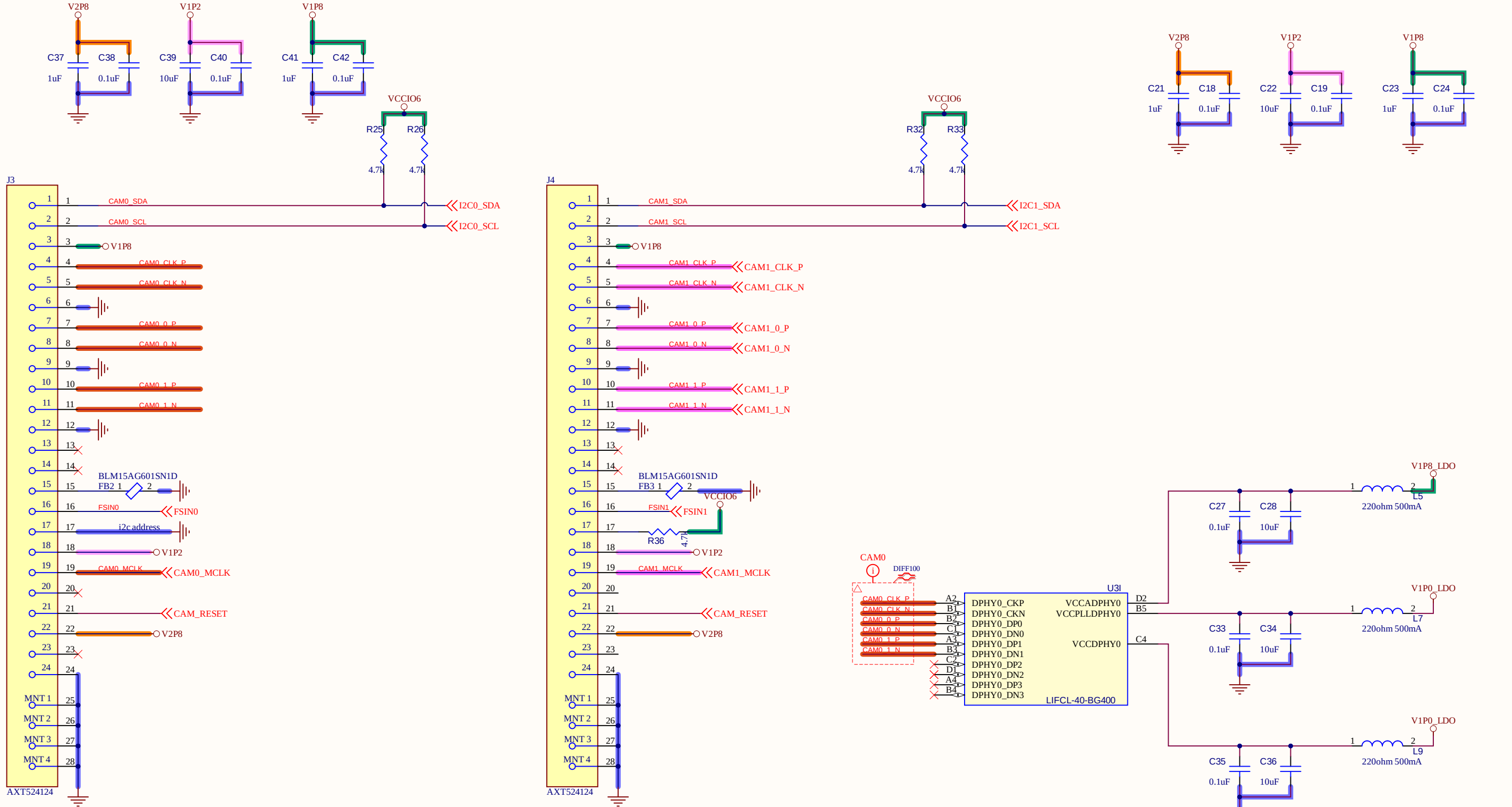


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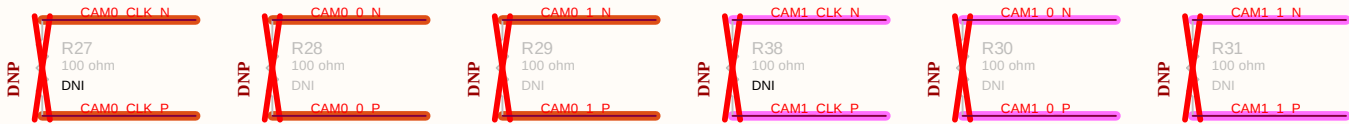
POWER DECOUPLING



CAMERA INTERFACE



LVDS RX Termination Resistors



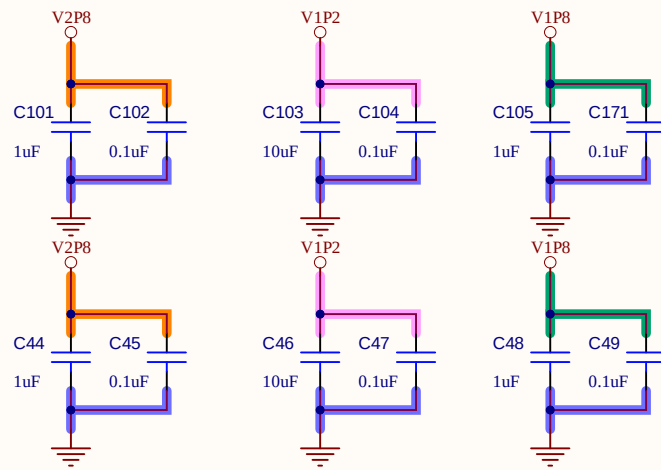
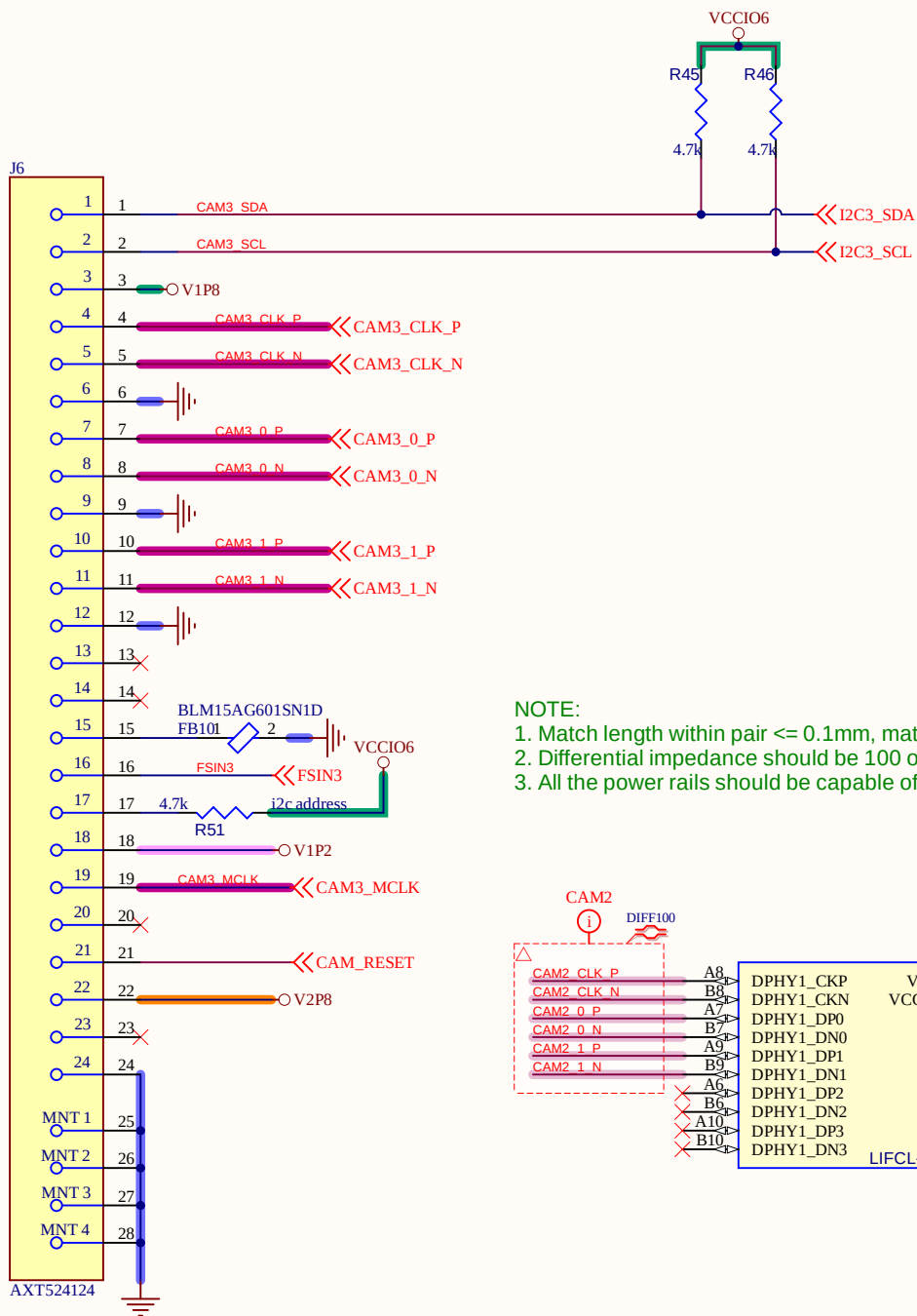
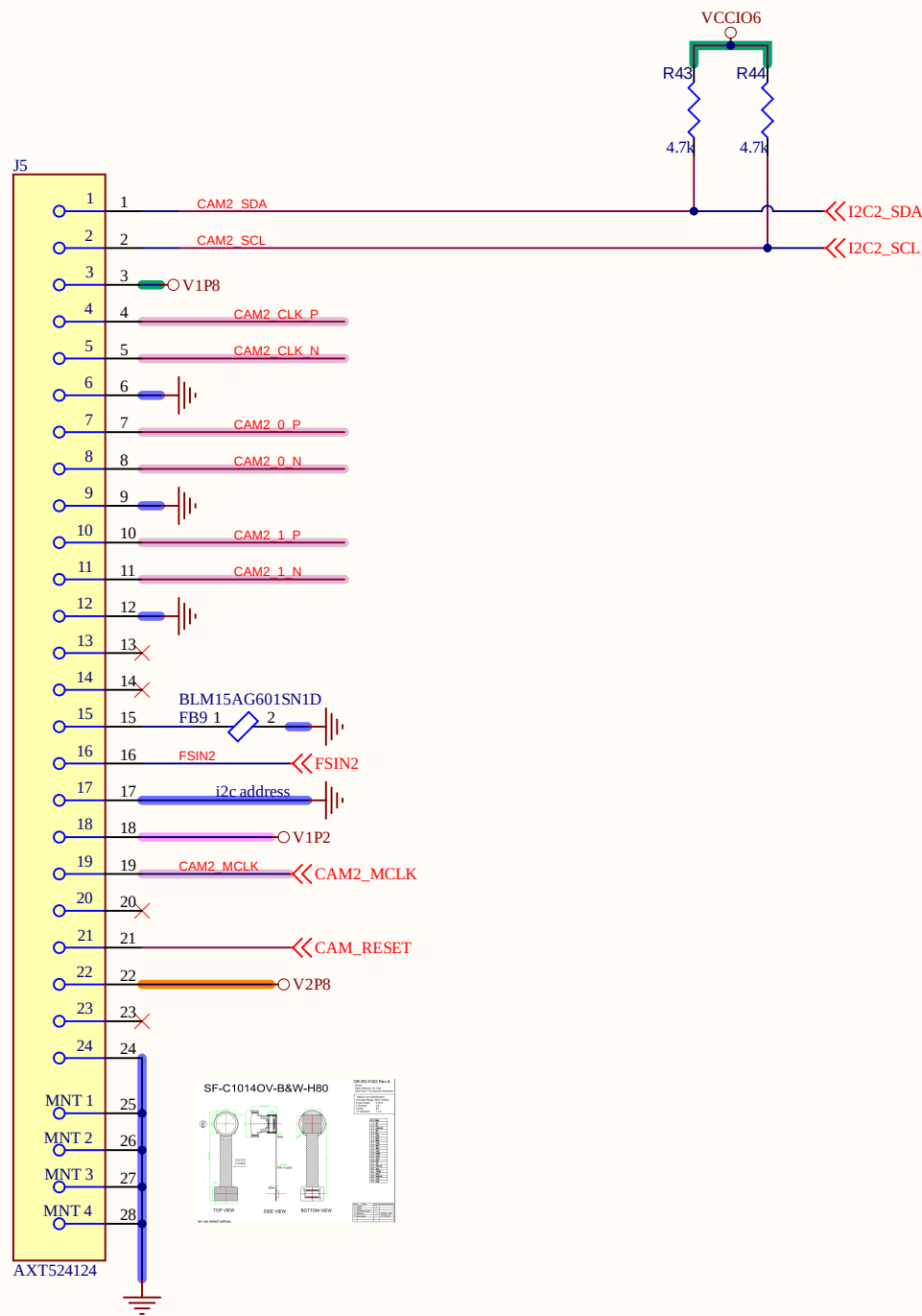
NOTE: Place close to FPGA

NOTE:

1. Match length within pair <= 0.1mm, match length between pairs <=1.0mm.
2. Differential impedance should be 100 ohms and 50 ohms as a single ended signal
3. All the power rails should be capable of carrying 1A current

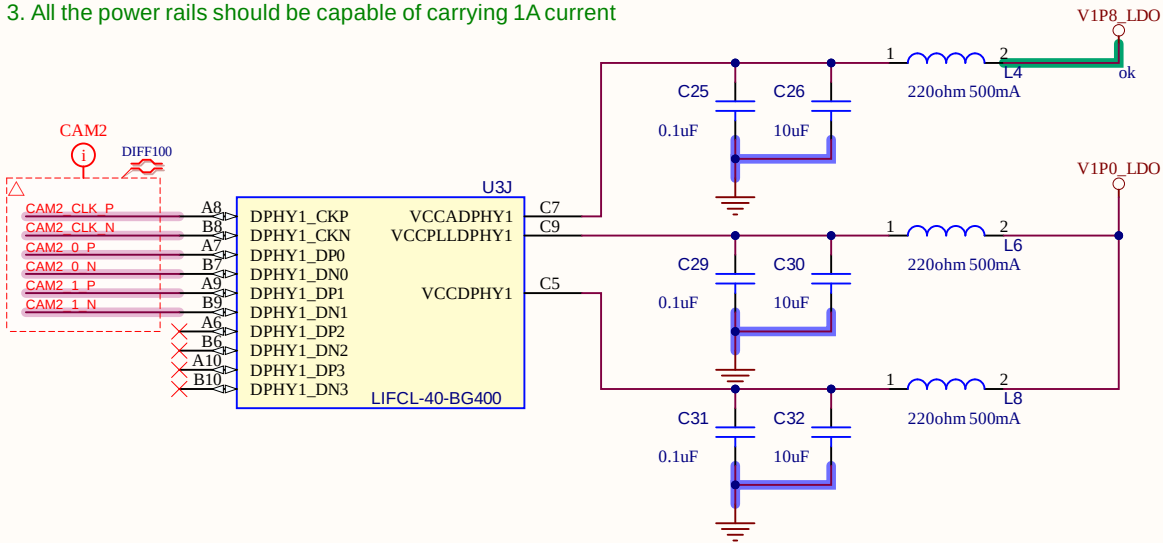
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	*	Umit KAYACIK	Revision: *
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	24.02.2026	*	Dil/Language: ENG
		Onaylayan/Approved By	Varyant/Variant:
		*	Variant of HW_CHRONOS_R1

CAMERA INTERFACE 2

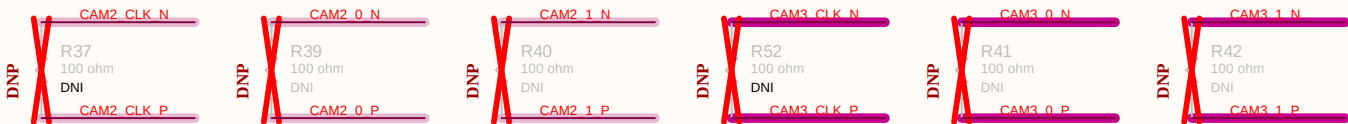


NOTE:

1. Match length within pair $\leq 0.1\text{mm}$, match length between pairs $\leq 1.0\text{mm}$.
2. Differential impedance should be 100 ohms and 50 ohms as a single ended signal
3. All the power rails should be capable of carrying 1A current



LVDS RX Termination Resistors



NOTE: Place close to FPGA

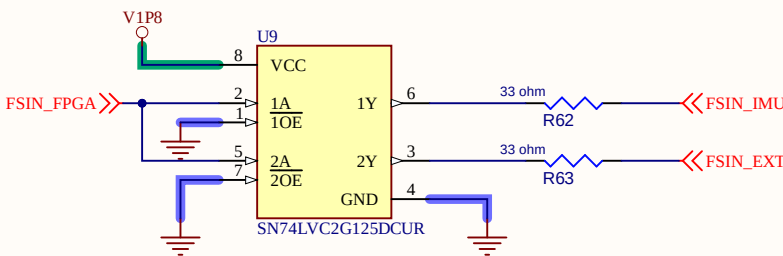
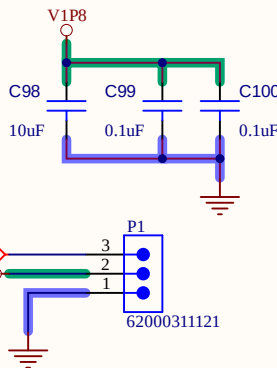
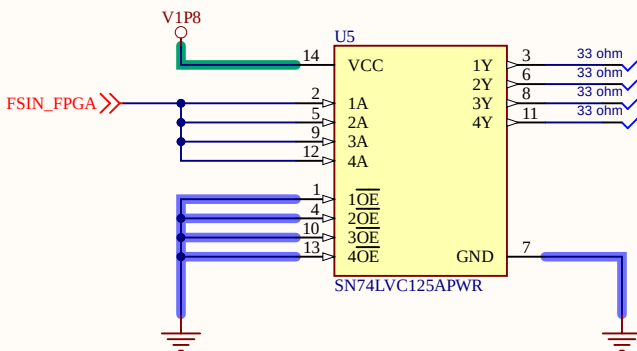
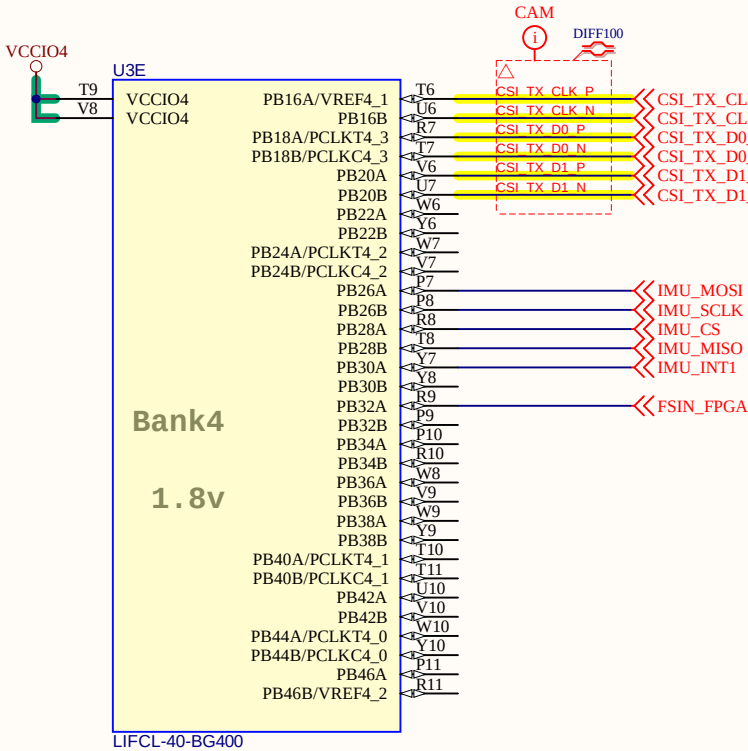
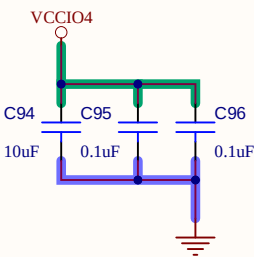
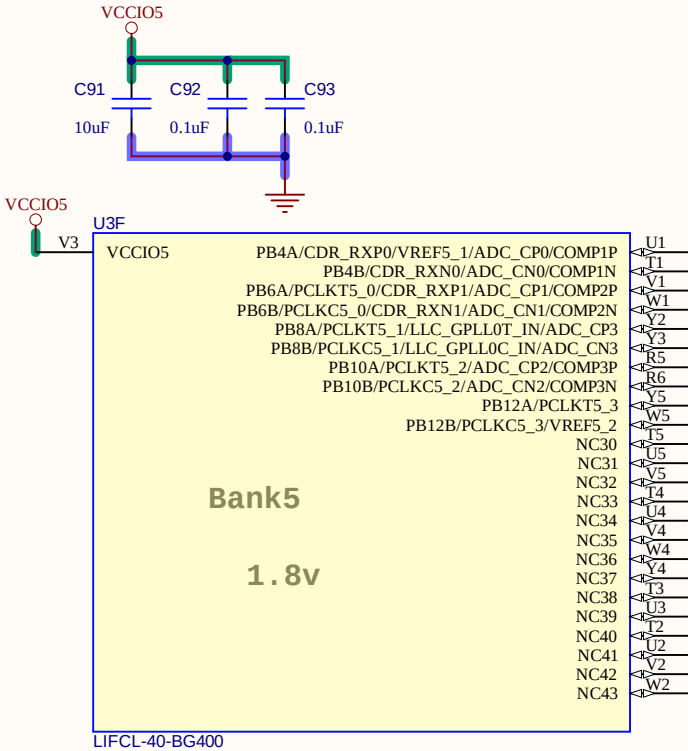
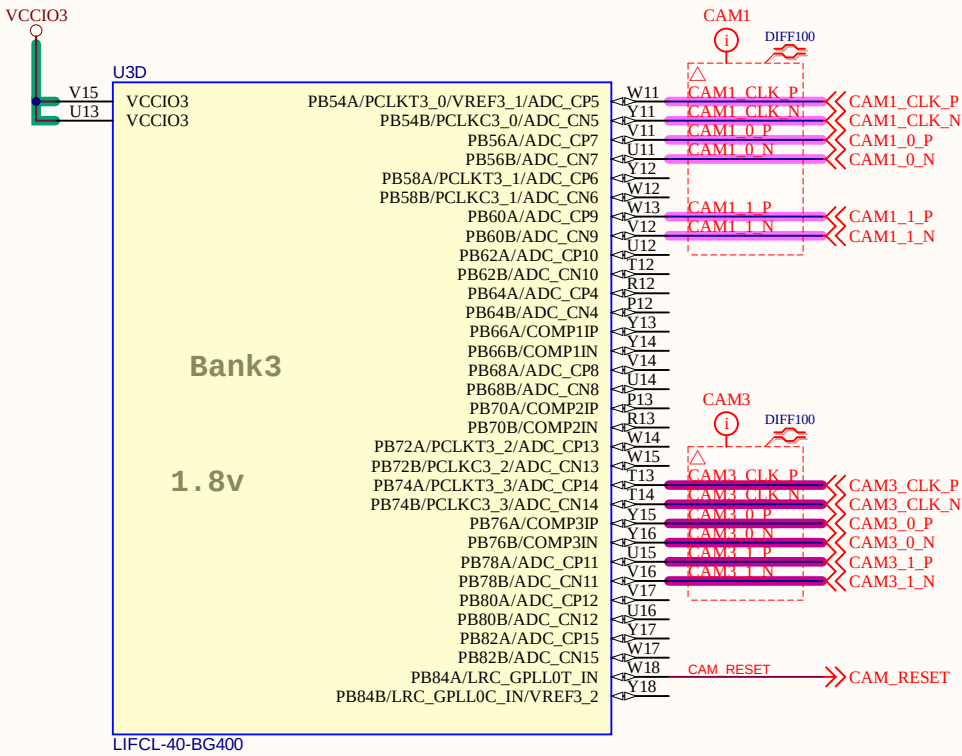
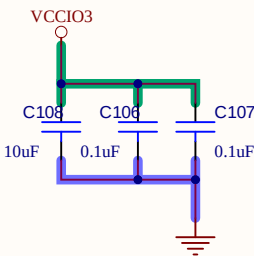
Keep LEDs away from Camera

Proje/Project: **HW_CHRONOS_R1**

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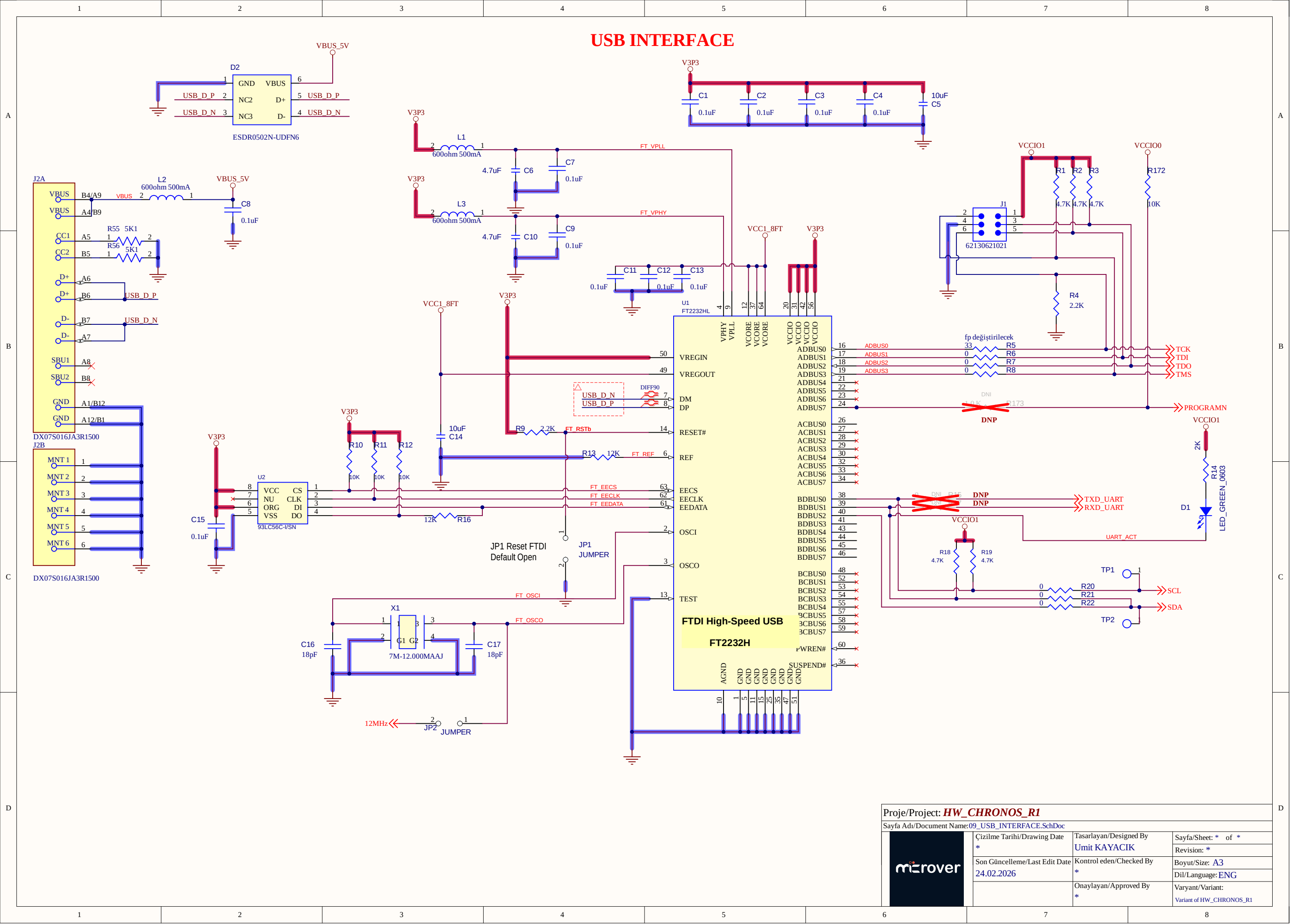
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	24.02.2026	*	Boyut/Size: A3
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		*	Varyant/Variant: Variant of HW_CHRONOS_R1

SOFT DPHY



Proje/Project: HW_CHRONOS_R1			
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	24.02.2026	*	Dil/Language: ENG
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	*		Variant of HW_CHRONOS_R1

USB INTERFACE

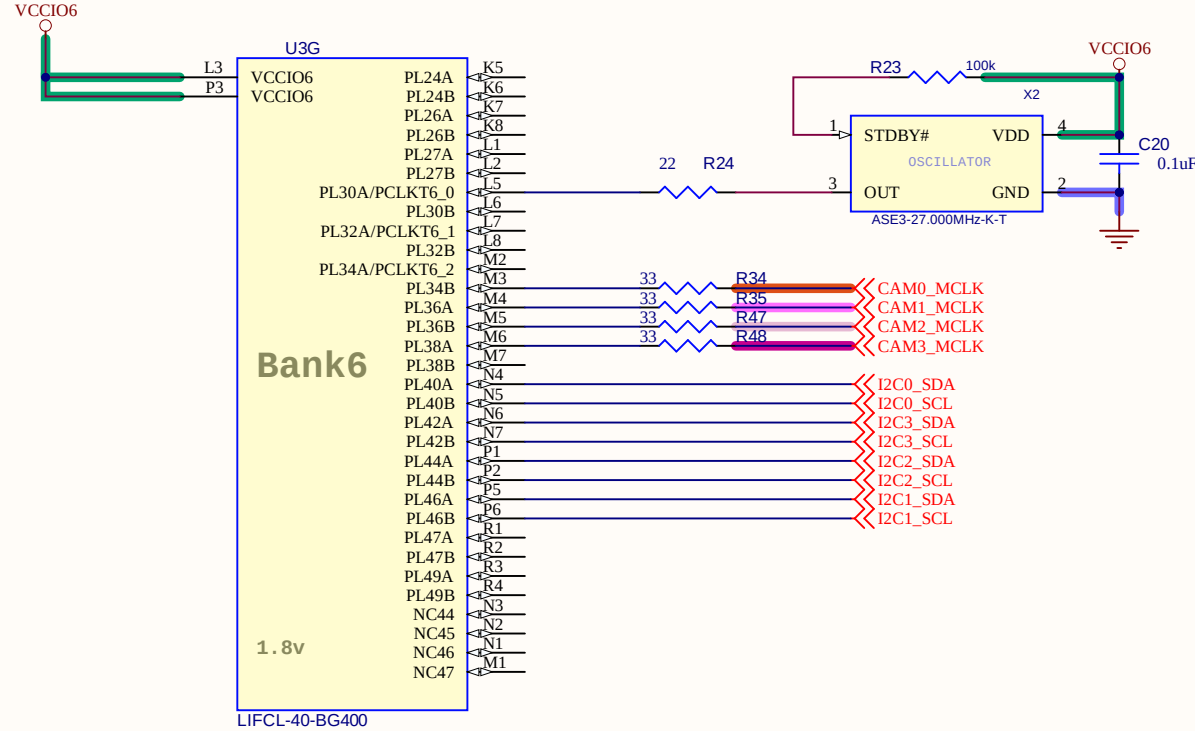
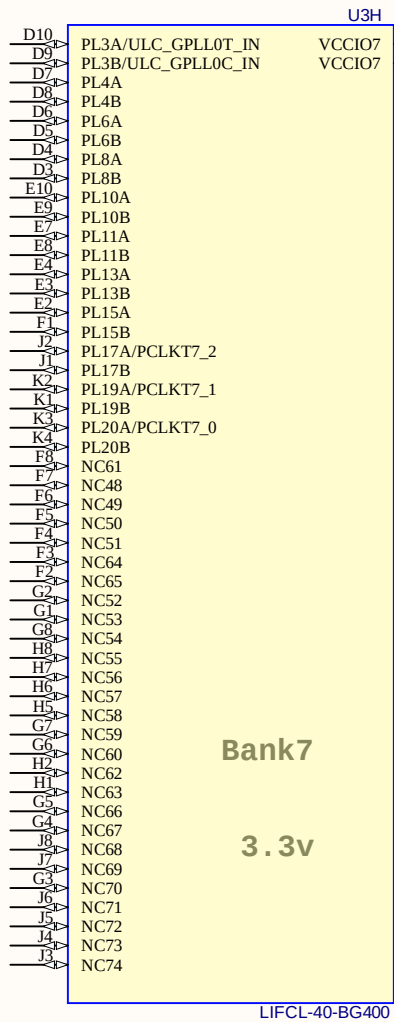
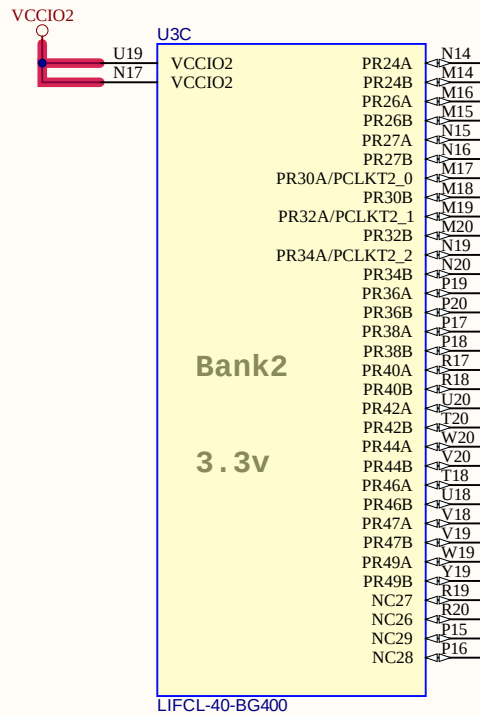
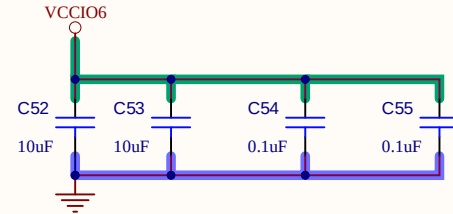
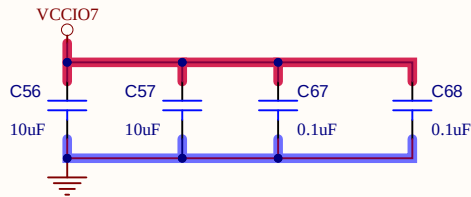
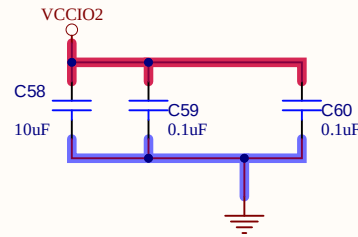


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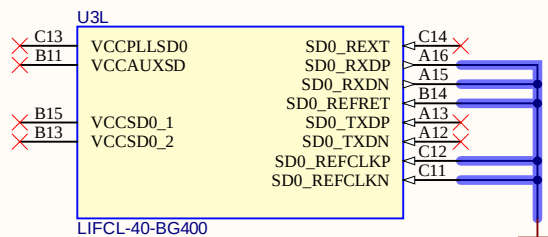
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	24.02.2026	*	Dil/Language: ENG
		Onaylayan/Approved By	Varyant/Variant:
		*	Variant of HW_CHRONOS_R1

BANK 2-6-7

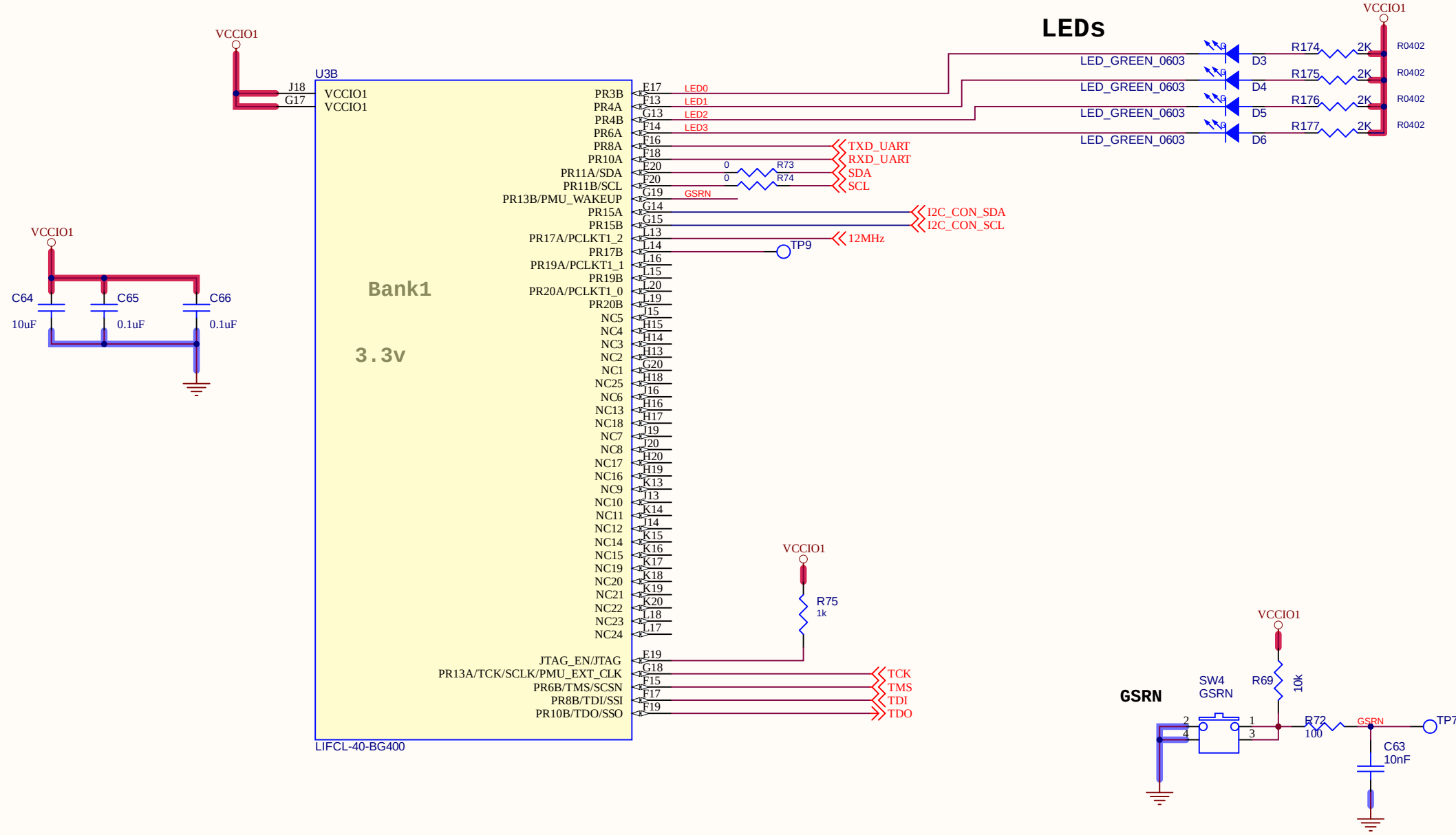


Route SMA pairs as 100 ohm differential



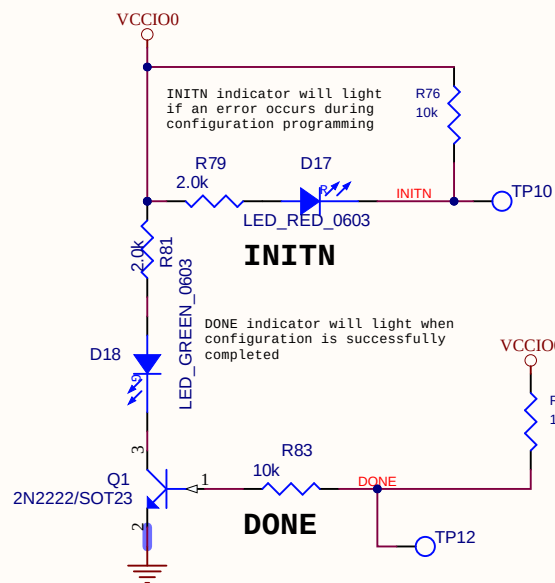
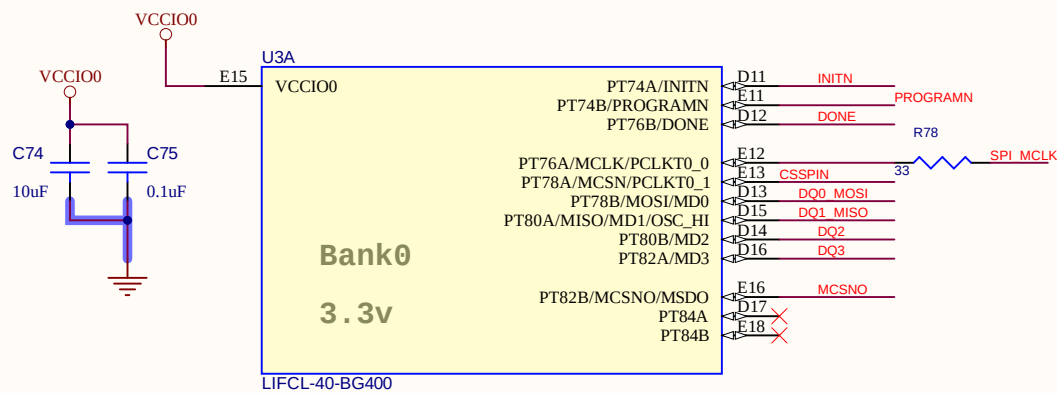
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PERIPHERALS

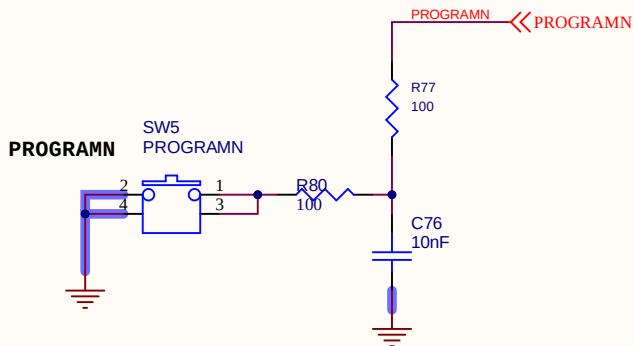
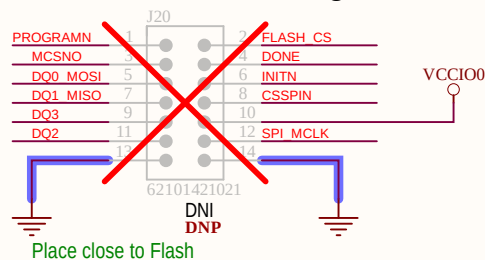


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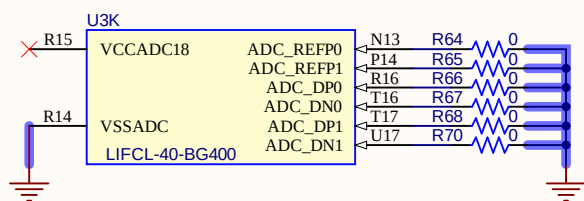
CONFIGURATION AND ADC



Parallel / SPI Config Header

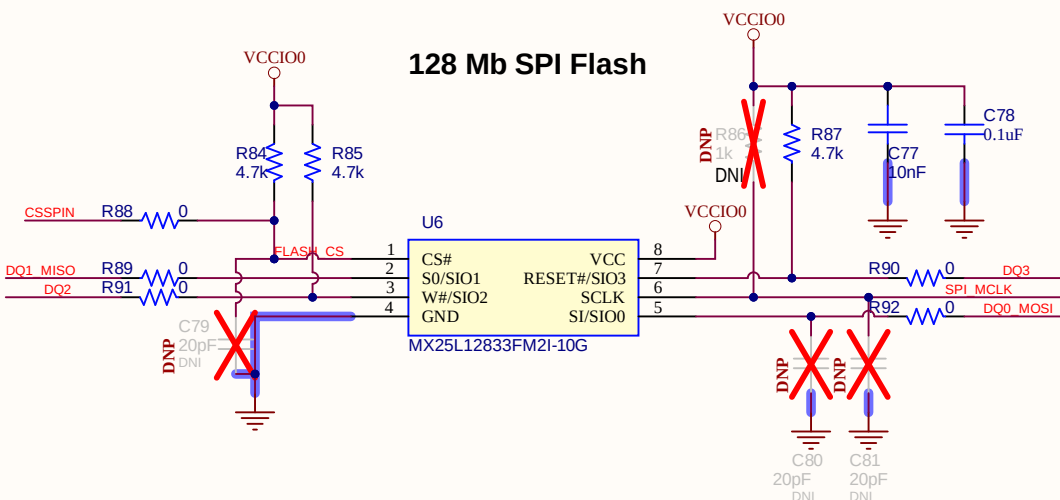


1.8v



Keep noisy signals away from ADC circuit

128 Mb SPI Flash

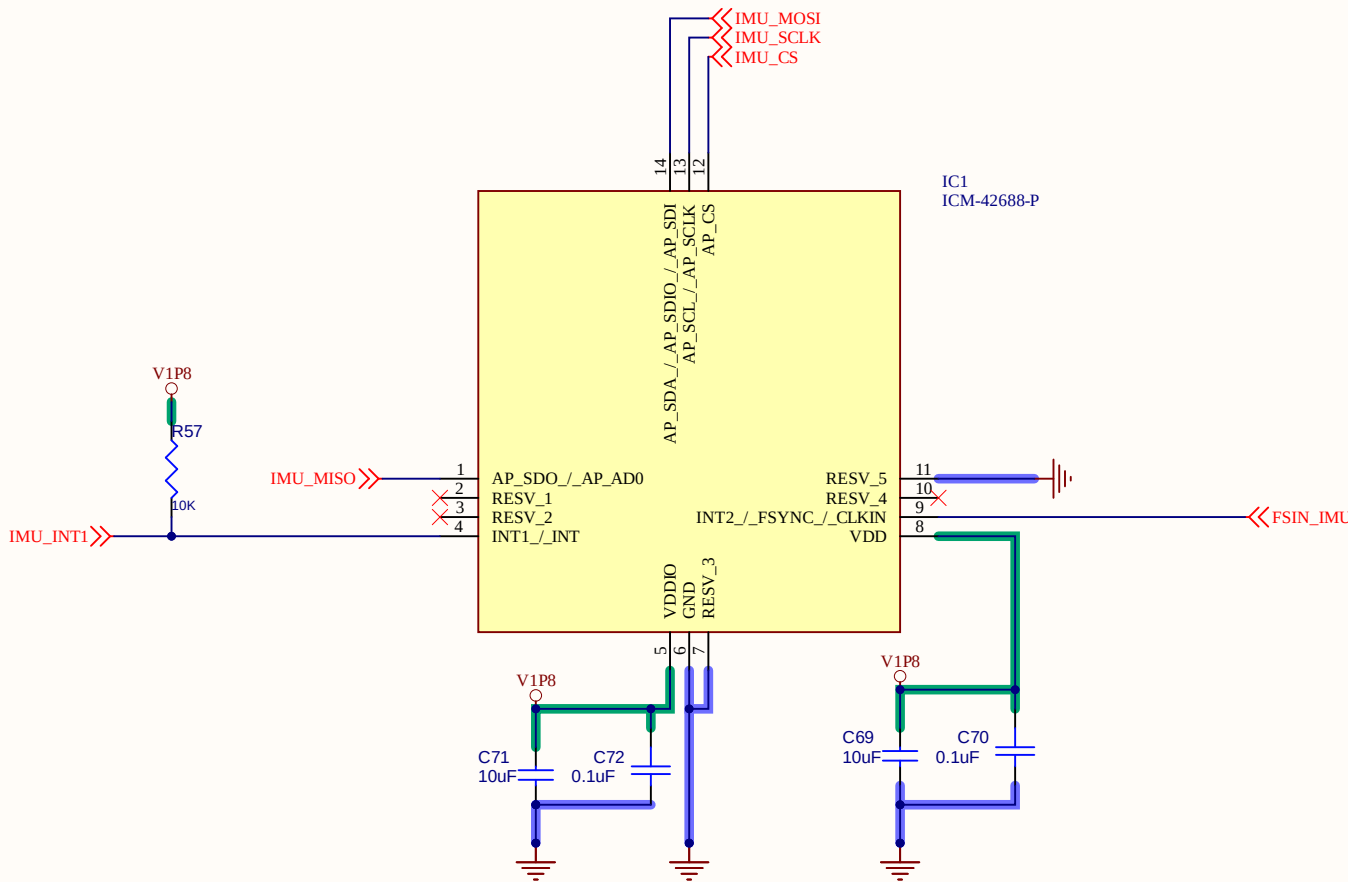


Proje/Project: **HW_CHRONOS_R1**

Sayfa Adı/Document Name: **12_CONFIGURATIN_AND_ADC.SchDoc**

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	24.02.2026	*	Dil/Language: ENG
		Onaylayan/Approved By	Varyant/Variant:
		*	Variant of HW_CHRONOS_R1

IMU



Proje/Project: **HW_CHRONOS_R1**

Sayfa Adı/Document Name: **13_IMU.SchDoc**

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	24.02.2026	*	Dil/Language: ENG
		Onaylayan/Approved By	Varyant/Variant:
		*	Variant of HW_CHRONOS_R1

JETSON CONNECTOR

